Mohammed Khadem, PhD

Doctor of Engineering in Robotics & Mechatronics
Former Research Fellow, Department of Mechanical, Energy and Management Engineering (DIMEG)
University of Calabria, Rende (CS), Italy

Contact Information

Address: Via in Rende 87036 (CS), Italy
Phone: +39 333 9446081
Email: mohammedkhadem1994@gmail.com | mohammed.khadem@unical.it
Nationality: Algerian
Date & Place of Birth: November 16, 1994 – Guemar, El-Oued, Algeria
Marital Status: Single

Research Profile

I am a Mechatronics Engineer and researcher with a PhD in Mechatronics Engineering from the University of Skikda, Algeria, and completed postdoctoral research experience at the University of Calabria, Italy. My research focuses on the design, modeling, simulation, prototyping, and control of advanced robotic systems, including cable-driven parallel robots, drones, mobile robots, and rehabilitation robotics. My expertise combines mechanical design, robotics, control systems, and software development, with experience in experimental validation and multidisciplinary international research projects. I have published research in international journals and conferences and continue to work on intelligent robotics, UAVs, artificial intelligence, and autonomous robotic systems.

❖ Research Interests

- Cable-Driven Parallel Robots (CDPRs)
- Rehabilitation Robotics
- Drone Design and Control
- Robot Modeling and Simulation
- Trajectory Planning and Control
- Mechanical Design for Robotics
- Cultural Heritage Inspection Robotics
- Human–Robot Interaction

❖ Academic Appointments

Postdoctoral Research Fellow

University of Calabria, DIMEG – Rende (CS), Italy
May 2024 – Abril 2026.

- Prototyping and development of the DIMEG heritage inspection drone
- Mechanical design using SolidWorks
- Simulation and control development using MATLAB
- Embedded programming (Arduino IDE, C/C++)
- Development of graphical user interfaces (Python, LabVIEW, MIT App Inventor, RemotXY)
- Experimental testing and data analysis

- Preparation of technical reports and scientific publications
- Participation in international conference

Project: *Tech4You – Technologies for Climate Change Adaptation and Quality of Life Improvement*
Supervisor: Prof. Giuseppe Carbone
Affiliation: DIMEG, University of Calabria

Temporary Assistant Professor

University of August 20, 1955 – Skikda, Algeria
November 2019 – June 2023

- **2019–2020:** Laboratory instruction in Microprocessor Systems
- **2021–2022:** Supervisory assistant for Master's students in two subjects
- **2022–2023:**
    - Teaching CFAO (CAD/CAM) with laboratory activities
    - Teaching Industrial Robotics with CFAO-related laboratory sessions
    - Supervisory assistant for Master's students in six robotics-related subjects

---

❖ Education

PhD in Mechatronics Engineering

University of August 20, 1955 – Skikda, Algeria
November 2019 – June 2024
Date of Award: September 10, 2024
Grade: Very Honorable

Dissertation Title:
Design and Control of a Cable-Driven Parallel Robot with Rehabilitation Application

Supervisor: Prof. Fouad Inel, Mechanical Engineering Department, University of Skikda
Co-supervisor: Prof. Giuseppe Carbone, DIMEG, University of Calabria

Research Activities:

- Design and prototyping of a pyramidal CDPR
- Kinematic and dynamic simulation (MATLAB)
- Mechanical design (SolidWorks)
- Embedded programming (Arduino IDE, C/C++)
- Interface development (LabVIEW, MIT App Inventor, and Python)
- PCB design, prototyping, and manufacturing
- Scientific publishing and conference participation

Master's Degree in Mechatronics Engineering

University of August 20, 1955 – Skikda, Algeria
October 2017 – June 2019
Date of Award: July 28, 2019
Grade: **Valedictorian**

Bachelor's Degree in Electromechanical Engineering

University of El Oued – Martyr Hama Lakhdar, Algeria
September 2014 – June 2017
Date of Award: June 28, 2017
Grade: Good

❖ International Training

Short-Term Advanced Training Course (Robotics & Mechatronics)

University of Calabria, Rende (CS), Italy
January 19 – February 16, 2023
Focus on robotics, mechatronics, and international research collaboration

❖ Publications

1. Khadem, M., Inel, F., Carbone, G., & Slimane Tich Tich, A. (2023). *A novel pyramidal cable-driven robot for exercising and rehabilitation of writing tasks.* Robotica, 1–22. https://doi.org/10.1017/S026357472300111X
2. Khadem, M., Inel, F., & Carbone, G. (2022). *Design and Validation of a Novel Pyramidal Cable-Driven Robot.* IFToMM Conference, Springer. https://doi.org/10.1007/978-3-031-10776-4_24
3. Khadem, M., Inel, F., & Carbone, G. (2022). *Simulation of 2D paths using a Cable-Driven Parallel Robot.* In conference at: ICEE2022, Skikda. Algeria.
4. Khadem, M., Inel, F., & Carbone, G. (2022). *Simulation of Predefined Trajectories for a 3D Cable-Driven Parallel Robot.* In conference at: ICEE2022, Skikda. Algeria.
5. Inel, F., Khadem, M., & Slimane Tich Tich, A. (2022). *Realization and Simulation of a Novel Parallel Cable-Based Robot with Five Cables.* WSEAS Transactions on Circuits and Systems, 21. https://doi.org/10.37394/23201.2022.21.30
6. Tich, A. S. T., Inel, F., & Khadem, M. (2022). *Realization and Control of a Mobile Robot Using Artificial Intelligence.* WSEAS Transactions on Systems, 21. https://doi.org/10.37394/23202.2022.21.39
7. Sista, A. A., Assas, N., & Khadem, M. (2023). *PID Control Applied to the Simulation of a Manipulator Arm Using LabVIEW.* In conference at: ICPET23, Skikda. Algeria
8. Khadem, M., Inel, F., & Carbone, G. (2023). *Design of a 3D Cable-Driven Robot for Material Transmission in a Petrochemical Base.* In conference at: ICPET23. Skikda. Algeria.
9. Inel, F., Khadem, M., & Slimane Tich Tich, A. (2024). *Objective-Oriented Design of a CUBE Cable-Based Parallel Robot for Arm Rehabilitation.* WSEAS Transactions on Systems, 23. https://doi.org/10.37394/23202.2024.23.1
10. Khadem, M., Malyshev, D., & Carbone, G. (2024). *Proposed Design of New Drones for Cultural Heritage Applications.* IFToMM YFG-MMS 24. https://doi.org/10.17185/duepublico/82582
11. Khadem, M., Malyshev, D., Leone, S., & Carbone, G. (2025). *A Design Procedure Towards a Drone for Coastal Cultural Heritage Inspection.* I4SDG 2025, Springer. https://doi.org/10.1007/978-3-031-91179-8_1
12. Khadem, M., et al. (2025). *Preliminary Experimental Test of Thermal Exploration for Cultural Heritage Applications.* Jc-IFToMM International Symposium. https://doi.org/10.57272/jciftomm.8.0_71

13. Khadem, M. & Carbone, G. (2025). “Performance Assessment of the DIMEG Heritage Drone Using Real-Time Control and Stabilization Mode” Conference: 2025 4th International Conference on Advanced Control, Automation and Robotics (ICACAR 2025). At: Cassino. Italy. https://doi.org/10.13140/RG.2.2.23307.35364

14. Khadem, M. & Carbone, G. (2026). “Conceptual Design of a Pyramidal Cable-Driven Rehabilitation Robot with Sliding Actuation for Upper Limb Assistance” In book: Advances in Motion Assisting Devices for Limbs. https://doi.org/10.1007/978-3-032-15271-8_10

15. Carbone, Giuseppe, Mohammed Khadem, and Dmitry Malyshev. "Comprehensive Review and Systematic Design Approach for Drones Employed in Inspection and Preservation of Built Cultural Heritage Sites." *Robotics and Autonomous Systems* (2026): 105407. https://doi.org/10.1016/j.robot.2026.105407

16. Mohammed Khadem, Dmitry Malyshev, and Giuseppe Carbone. 2025. Automatic balancing control experiment on an axis for heritage drone prototype using PID control. In Advances in Robotics 2025 (AIR 2025), July 02–05, 2025, Jodhpur, India. ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3787370.3787376

17. ABDELKRIM, M.; GUIA, H.; AROUDJ, M.; Khadem, M.; Carbone, G.; LAOUID, A. Design and Implementation of an SDM630-Based Energy Monitoring System for Three-Phase Electrical Machine Applications, in Proceedings of the 3rd International Electronic Conference on Machines and Applications, 12–14 May 2026, MDPI: Basel, Switzerland, https://sciforum.net/paper/view/30860

18. Aroudj, M.; Guia, H.; Khadem, M.; Mohrem, A.; Carbone, G. Comparative Performance Analysis of Spur and Chevron Gears in Wind Turbine Applications, in Proceedings of the 3rd International Electronic Conference on Machines and Applications, 12–14 May 2026, MDPI: Basel, Switzerland, https://sciforum.net/paper/view/30920

19. Khadem, M.; Leone, S.; Carbone, G. Adaptive Shield Mechanism for UAVs Ensuring Reliable Structural Performance During Thermal Monitoring of Heritage Sites, in Proceedings of the 3rd International Electronic Conference on Machines and Applications, 12–14 May 2026, MDPI: Basel, Switzerland, https://sciforum.net/paper/view/30940

20. Khadem, M.; Carbone, G.; Aroudj, M.; Guia, H. Proposed design of a cube cable-driven parallel robot design for rehabilitation exercises and precision 3D manufacturing, in Proceedings of the 3rd International Electronic Conference on Machines and Applications, 12–14 May 2026, MDPI: Basel, Switzerland, https://sciforum.net/paper/view/30940

21. Khadem, M.; Carbone, G (2026). *Advanced Technologies for the Inspection and Thermal Monitoring of Heritage Buildings Using Innovative DIMEG Heritage Drone.* Conference: Higher Education: New Trends & Technologies Webinar held on 5 May, Bahrain Polytechnic. https://doi.org/10.13140/RG.2.2.29632.29445

❖ Technical Skills

- **Software:** SolidWorks, MATLAB, LabVIEW, Python, Arduino IDE
- **Robotics:** Modeling, control, simulation, prototyping
- **Manufacturing:** 3D printing, PCB design
- **Data & Reporting:** Microsoft Word, Excel, PowerPoint

❖ Languages

- **Arabic:** Native
- **English:** B1 (Reading: Good | Writing: Good | Speaking: Good)
- **Italian:** beginner A1

❖ Academic & Professional Service

- Supervision and assistance of Master's students
- Organization of training courses in programming, simulation, CAD, and robotics
- Active participation in international conferences

❖ Additional Skills & Interests

- Technical drawing and artistic sketching
- Multicultural teamwork and academic collaboration

❖ Online Profiles

- ORCID
- Google Scholar
- ResearchGate
- Linked in
- YouTube channel
